

A Formal Taxonomy of Physical Resonance Mechanisms

Oldřich Dvořák*
Independent researcher
(Dated: 2026-05-24)

We propose a formal taxonomy of physical resonance mechanisms keyed to the load-bearing mathematical object rather than to broad community naming conventions or surface morphology. Within the tested scope, the best top-level split is a four-family taxonomy: **F1** closed/Hermitian spectral resonance, **F2** open/non-Hermitian pole or quasinormal-mode resonance, **F3** nonlinear phase-locking resonance, and **F4** Floquet/parametric resonance. Admission into the taxonomy requires five fields for each row: governing object, resonance condition, observable, parameter packet, and demotion test. Cross-domain transfer is graded by an EXACT/ANALOGY gate: EXACT requires the same reduced operator structure or invariant packet up to named boundary conditions, whereas ANALOGY permits behavioral resemblance without mechanism identity. Across 54 documented case-tests in the public supplement, including classical, quantum, open-system, nonlinear, driven, many-body, noise-activated, delayed-memory, fractional/nonlocal, non-Markovian, active-matter, and time-crystal examples, we find no fifth-family survivor. This is a bounded claim about the audited corpus, not an unbounded closure claim.

I. PROBLEM STATEMENT

The literature uses the word resonance for sharply different physical mechanisms: discrete mode excitation, scattering poles, synchronization, parametric instability, interference line shapes, exceptional points, bound states in the continuum, stochastic rate matching, and discrete time-crystal response. A usable taxonomy cannot be built from vocabulary alone. It must classify systems by the mathematical object without which the phenomenon ceases to be the same phenomenon.

The central claim of this paper is therefore taxonomic rather than metaphysical. We do not claim that all future resonance-like phenomena must fall into our scheme. We claim only that, under the current corpus and admissibility contract, the four families below absorb all admitted cases and leave zero confirmed pressure for a fifth top-level family.

II. ADMISSIBILITY CONTRACT

Any row admitted to the taxonomy must specify five fields:

1. **Governing object:** the load-bearing mathematical object that organizes the phenomenon.
2. **Resonance condition:** the condition under which the effect is expected to occur.
3. **Observable:** the measurable response class.
4. **Parameter packet:** the minimal control and system parameters required to state the prediction.

5. **Demotion test:** the falsifier that would demote the row to analogy, adjacency, or non-resonant dynamics.

This five-field contract is operationalized by six invariants.

T1. Mechanism packet invariant

A resonance claim must name participating degrees of freedom, the relevant spectral or phase object, the coupling or drive, the damping or linewidth or locking range or instability width, and a measurable response variable. If one of those objects is missing, the claim is not a mechanism row.

T2. Family-membership invariant

A row belongs to a top-level family only by its governing mathematical object: closed spectral, open pole/QNM, nonlinear phase-locking, or Floquet/parametric. If the claimed family can be removed without destroying the phenomenon, the row is misclassified.

T3. Observable discriminator invariant

A resonance claim is testable only when it states the observable class expected to change: amplitude peak, absorbed power, linewidth, ringdown decay rate, phase lag, locking order parameter, instability exponent, transmission zero, or another explicit measurable.

* oldrich@oldrich.me

T4. Exact-transfer invariant

Cross-domain reuse is EXACT only when the reduced operator structure or invariant packet survives the transfer, including a named failure boundary. Similar plots or vocabulary alone do not constitute mechanism transfer.

T5. Selection-rule invariant

Selection-rule language is exact only when a coupling or projection operator is defined and its null or non-null structure enters the prediction. Otherwise selection-rule wording is merely heuristic.

T6. Open-system subfamily invariant

Exceptional points, Fano structures, bound states in the continuum, dark states, and related zero/interference phenomena are subfamily or observable-level statements inside the open family. They are not independent top-level families unless they introduce a distinct governing object.

III. THE FOUR TOP-LEVEL FAMILIES

A. F1: closed/Hermitian spectral resonance

The defining object of F1 is a self-adjoint or effectively closed spectral problem with discrete modes or gaps. Canonical examples include molecular-beam magnetic resonance, bulk NMR, and cyclotron resonance. The decisive signature is frequency-selective excitation of a discrete mode structure with an explicit resonance observable and a demotion test that excludes broadband heating or non-specific response.

B. F2: open/non-Hermitian pole or quasinormal-mode resonance

The defining object of F2 is a complex pole, outgoing-boundary eigenmode, resolvent singularity, or equivalent open-system spectral object. Canonical examples include Breit-Wigner nuclear resonance and black-hole quasinormal modes. This family also hosts boundary packets and observable overlays such as exceptional points, Fano interference, and bound states in the continuum, provided their open-system singular or embedded-state structure is made explicit. If the effect can be stated without a complex pole, outgoing-boundary mode, or equivalent open-system spectral object, the row demotes to analogy, adjacency, or a different family.

C. F3: nonlinear phase-locking resonance

The defining object of F3 is a reduced nonlinear phase dynamics with explicit detuning, coupling, and a lock or unlock boundary. Canonical examples include Adler injection locking, nonlinear subharmonic entrainment, and Shapiro-step Josephson locking. A peak without phase capture is not enough.

D. F4: Floquet/parametric resonance

The defining object of F4 is a periodically modulated operator, Floquet map, monodromy operator, quasienergy structure, or instability tongue generated by coefficient modulation or periodic driving. Canonical examples include Mathieu-type parametric resonance, Faraday waves, and periodically driven many-body systems where the governing object is explicitly Floquet. If periodic driving does not supply a load-bearing Floquet, monodromy, quasienergy, or instability-tongue object, the row demotes to analogy, adjacency, or a different family.

IV. EXACT VERSUS ANALOGY

The taxonomy is not only a classification scheme but a transfer-discipline rule.

EXACT means the same reduced operator structure or invariant packet survives the translation between domains, together with a named failure boundary. For example, a pole-governed ringdown in gravitational physics and a pole-governed scattering resonance in another open system may legitimately share F2 structure if the pole description does the explanatory work in both cases.

ANALOGY means the behavior is similar but the governing object changes. Similar line shapes, instability onsets, or synchronization motifs are insufficient for exact transfer unless the operator-level reduction survives. This gate prevents the taxonomy from collapsing into vocabulary reuse.

V. BOUNDARY CASES AND ADJACENT NAMED PHENOMENA

The four-family structure only remains stable if common naming traps are handled explicitly.

A. Exceptional points, Fano, and BIC

Exceptional points are not a fifth resonance family. They are a strict open-family boundary packet requiring a defective complex-spectral singularity plus topology evidence. Fano structures are observable overlays inside

F2, not top-level families. Bound states in the continuum are embedded-state boundary cases inside F2, not independent parents.

B. Stochastic, coherence, and vibrational resonance

Stochastic resonance, coherence resonance, and vibrational resonance remain adjacent-but-not-admitted under the present taxonomy. These are real phenomena, but the current audited rows do not share a single governing object strong enough to force a fifth deterministic top-level family. The correct bounded claim is that they remain a **noise-activated and vibration-assisted rate-matching** adjacent cluster.

C. Floquet discrete time crystals

Floquet discrete time crystals (DTCs) belong at the F4 boundary when classification is keyed to the governing mathematical object: the Floquet operator. The subharmonic response mechanism, including many-body eigenstate order, differs from standard parametric frequency selection but does not introduce a new governing-object class; therefore it does not justify a fifth family under this taxonomy.

This boundary sentence matters because DTCs are one of the strongest apparent challenges to the four-family split. The taxonomy survives only if classification is anchored to governing object rather than to the sociological label attached to the mechanism.

The publication-facing DTC source chain used for this boundary classification is the review/framework note of Sacha and Zakrzewski [1], the earlier modeling note of Sacha [2], the companion Floquet-time-crystal theory papers of Khemani et al. [3] and Else et al. [4], the Yao et al. theory anchor [5], the dissipative theory row of Lledo and Szymanska [6], and the dissipative observation of Kessler et al. [7].

VI. CORPUS RESULT

The publication corpus used here is the 54-case-test supplement ledger released with this manuscript as `2026-05-24_RF-PUB-13_54_case_public_supplement.tsv`. It materializes the audit count inherited from RF-TAX-04, RF-TAX-05, RF-TAX-06, and RF-TAX-07: 15 representative stress-test rows, 3 adjacent-rate-matching tribunal rows, 12 non-equilibrium counterexample rows, and 24 delayed/fractional/non-Markovian/active-matter/time-crystal counterexample rows. The appendix below prints a representative 15-row subset so that the five-field contract is visible in paper form; the supplement is the count-bearing ledger.

Representative audited boundary packets within those sectors include chaotic-system resonances [8], quantum-

chaotic phase-space structures [9], active-matter long-range order [10], and ghost resonance in heterogeneous neuronal pools [11].

Under the 54 case-tests in the public supplement, the result is negative discipline rather than taxonomy inflation. We find no fifth-family survivor in the stress-test ledger. The load-bearing decision rule is the governing object. Once that rule is enforced, many tempting extra families demote to subfamily, overlay, adjacent cluster, or analogy.

VII. SCOPE AND CLAIM DISCIPLINE

This paper makes a bounded claim. It does not assert exhaustive closure over every possible future resonance phenomenon or every broader framework in physics. It states that, for the 54 documented case-tests in the public supplement and the admissibility contract, the best top-level classification is four-family and that no tested case forces a fifth family.

That scope discipline is not rhetorical caution. It is structurally necessary. A taxonomy paper should overclaim only when it can name the mechanism that breaks the demotion rule. At present, no such fifth-family mechanism survives the audited corpus.

VIII. CONCLUSION

The publication-safe core of the taxonomy is narrow and defensible: resonance mechanisms are best classified by governing object; a five-field admissibility contract blocks morphology-driven drift; cross-domain transfer must clear an EXACT/ANALOGY gate; Floquet DTC remains an F4 boundary case; and the 54 documented case-tests in the public supplement yield no fifth-family survivor.

Appendix A: Representative Printed Stress-Test Ledger

This appendix prints a representative subset of the stress-test corpus used for the publication claim. The count-bearing 54-case-test ledger is the public supplement file `2026-05-24_RF-PUB-13_54_case_public_supplement.tsv`. Each row below is printed with the five admissibility fields from Sec. II: governing object, resonance condition, observable, parameter packet, and demotion test. The family label is downstream of the governing object.

1. Rabi molecular-beam magnetic resonance: PASS_EXACT as F1. Object: discrete Zeeman/Larmor spectral gap. Condition: RF drive near Larmor frequency. Observable: transition probability or beam-intensity dip. Packet:

- species, static field, RF amplitude/frequency, transit time, linewidth. Demotion: broadband heating or deflection without resolved gap [12].
2. Bulk NMR in solids: **PASS_EXACT** as F1. Object: nuclear spin splitting in a static field. Condition: RF field tuned to nuclear precession. Observable: absorption or induction signal. Packet: sample, field, RF frequency, coil geometry, relaxation/linewidth. Demotion: thermal response without frequency-selective spin transition [13].
 3. Cyclotron resonance in Si/Ge: **PASS_EXACT** as F1. Object: carrier cyclotron frequency in a magnetic field. Condition: microwave frequency near cyclotron frequency. Observable: microwave absorption versus field/orientation. Packet: material, carrier type, temperature, field strength/orientation, polarization, linewidth. Demotion: Drude rolloff or missing anisotropic cyclotron scaling [14].
 4. Breit-Wigner nuclear resonance: **PASS_EXACT** as F2. Object: isolated complex S-matrix/resolvent pole. Condition: scattering energy near the pole. Observable: cross-section peak and phase-shift behavior. Packet: resonance energy, total width, partial widths, channel angular momentum, background. Demotion: threshold cusp or smooth background artifact [15].
 5. Black-hole quasinormal modes: **PASS_EXACT** as F2. Object: complex QNM frequencies with outgoing/ingoing boundary conditions. Condition: perturbation projects onto damped QNM spectrum. Observable: ringdown frequencies and damping times. Packet: mass, spin, mode indices, start-time choice, damping time, signal-to-noise. Demotion: generic damped sinusoid without QNM boundary discipline [16].
 6. Exceptional point in a microwave cavity: **PASS_SPLIT_BOUNDARY** inside F2. Object: defective non-Hermitian spectral branch point. Condition: controls tune to an EP and a loop encircles it. Observable: eigenvalue/eigenvector coalescence and interchange. Packet: two controls, complex eigenvalues, loop geometry, field-pattern reconstruction. Demotion: generic non-normal gain/loss without coalescence and topology [17].
 7. Optical bound state in the continuum: **PASS_SPLIT_BOUNDARY** inside F2. Object: embedded eigenvalue or zero-leakage state in a radiation continuum. Condition: radiation amplitudes cancel while the mode remains embedded. Observable: trapped light or divergent Q . Packet: geometry, wave vector, symmetry/interference channel, leakage amplitudes, Q scaling. Demotion: ordinary finite- Q guided resonance [18].
 8. Fano autoionization line shape: **PASS_OBSERVABLE_OVERLAY** inside F2. Object: discrete-continuum interference. Condition: discrete autoionizing level overlaps a continuum. Observable: asymmetric line shape and phase shift. Packet: resonance energy, width, asymmetry parameter, continuum channel, background amplitude. Demotion: line-shape naming without pole-plus-continuum decomposition [19].
 9. Adler injection locking: **PASS_EXACT** as F3. Object: phase-reduced nonlinear oscillator. Condition: detuning lies inside coupling-defined lock range. Observable: frequency pulling, phase capture, lock/unlock transition. Packet: free-running frequency, injected frequency, coupling strength, damping, lock range. Demotion: linear response peak without phase capture or phase-slip boundary [20].
 10. Van der Pol frequency demultiplication: **PASS_EXACT** as F3. Object: self-oscillatory limit cycle with nonlinear phase locking. Condition: forcing falls inside a subharmonic locking tongue. Observable: demultiplied or entrained output frequency. Packet: forcing frequency/amplitude, autonomous frequency, nonlinear damping parameters. Demotion: passive harmonic response without autonomous phase dynamics [21].
 11. Shapiro steps in Josephson tunneling: **PASS_EXACT** as F3. Object: Josephson phase dynamics on a synchronization manifold. Condition: AC Josephson frequency locks to the microwave drive. Observable: constant-voltage steps in the current-voltage curve. Packet: RF frequency/power, DC bias, critical current/damping, step index. Demotion: cavity or transport features without Josephson phase locking [22].
 12. MEMS parametric-resonance tongues: **PASS_EXACT** as F4. Object: periodically modulated operator or Mathieu-Floquet monodromy. Condition: drive frequency and modulation depth cross an instability tongue. Observable: instability-lobe onset and parametrically excited oscillation. Packet: natural frequency, drive frequency, modulation amplitude, damping, order. Demotion: direct resonance without Floquet tongue structure [23].
 13. Faraday waves: **PASS_EXACT** as F4. Object: Floquet instability of a periodically forced free-surface mode. Condition: vertical forcing crosses subharmonic threshold. Observable: standing-wave onset and pattern selection. Packet: drive frequency, acceleration, fluid depth, viscosity, surface tension, wave number. Demotion: pattern formation without subharmonic threshold or mode-selection structure [24].

14. Discrete time crystal: **PASS_EXACT** at the **F4** boundary. Object: Floquet many-body evolution operator with robust subharmonic orbit. Condition: drive plus disorder/interactions stabilizes response at a multiple of the drive period. Observable: long-lived subharmonic temporal correlations. Packet: drive sequence/period, disorder, interaction scale, lifetime/robustness window. Demotion: transient or fine-tuned period doubling [1, 3–5, 25, 26].
15. Stochastic resonance: **QUARANTINE_ADJACENT**. Ob-

ject: noise-assisted switching or Kramers-rate matching, not one of the four deterministic top-level objects. Condition: noise tunes switching timescale to weak periodic forcing. Observable: signal-to-noise or spectral-power enhancement. Packet: barrier, forcing period/amplitude, noise intensity, switching-rate estimate. Demotion: keep adjacent unless a separate noise-activated rate-matching family is intentionally added [27].

-
- [1] K. Sacha and J. Zakrzewski, Time crystals: a review, *Reports on Progress in Physics* **81**, 016401 (2018).
- [2] K. Sacha, Modeling spontaneous breaking of time-translation symmetry, *Physical Review A* **91**, 033617 (2015).
- [3] V. Khemani, A. Lazarides, R. Moessner, and S. L. Sondhi, Phase structure of driven quantum systems, *Physical Review Letters* **116**, 250401 (2016).
- [4] D. V. Else, B. Bauer, and C. Nayak, Floquet time crystals, *Physical Review Letters* **117**, 090402 (2016).
- [5] N. Y. Yao, A. C. Potter, I.-D. Potirniche, and A. Vishwanath, Discrete time crystals: Rigidity, criticality, and realizations, *Physical Review Letters* **118**, 030401 (2017).
- [6] C. Lledo and M. H. Szymanska, A dissipative time crystal with or without z_2 symmetry breaking, *New Journal of Physics* **22**, 075002 (2020).
- [7] H. Kessler, P. Kongkhambut, C. Georges, L. Mathey, J. G. Cosme, and A. Hemmerich, Observation of a dissipative time crystal, *Physical Review Letters* **127**, 043602 (2021).
- [8] D. Ruelle, Resonances of chaotic dynamical systems, *Physical Review Letters* **56**, 405 (1986).
- [9] O. Bohigas, S. Tomsovic, and D. Ullmo, Manifestations of classical phase space structures in quantum mechanics, *Physics Reports* **223**, 43 (1993).
- [10] J. Toner and Y. Tu, Long-range order in a two-dimensional dynamical xy model: How birds fly together, *Physical Review Letters* **75**, 4326 (1995).
- [11] P. Balenzuela, J. Garcia-Ojalvo, E. Manjarrez, L. Martinez, and C. R. Mirasso, Ghost resonance in a pool of heterogeneous neurons, *Biosystems* **89**, 166 (2007).
- [12] I. I. Rabi, J. R. Zacharias, S. Millman, and P. Kusch, A new method of measuring nuclear magnetic moment, *Physical Review* **53**, 318 (1938).
- [13] E. M. Purcell, H. C. Torrey, and R. V. Pound, Resonance absorption by nuclear magnetic moments in a solid, *Physical Review* **69**, 37 (1946).
- [14] G. Dresselhaus, A. F. Kip, and C. Kittel, Cyclotron resonance of electrons and holes in silicon and germanium crystals, *Physical Review* **98**, 368 (1955).
- [15] E. P. Wigner, Resonance reactions and anomalous scattering, *Physical Review* **70**, 15 (1946).
- [16] M. Maggiore, Physical interpretation of the spectrum of black hole quasinormal modes, *Physical Review Letters* **100**, 141301 (2008).
- [17] C. Dembowski, H.-D. Gräf, H. L. Harney, A. Heine, W. D. Heiss, H. Rehfeld, and A. Richter, Experimental observation of the topological structure of exceptional points, *Physical Review Letters* **86**, 787 (2001).
- [18] C. W. Hsu, B. Zhen, J. Lee, S.-L. Chua, S. G. Johnson, J. D. Joannopoulos, and M. Soljačić, Observation of trapped light within the radiation continuum, *Nature* **499**, 188 (2013).
- [19] U. Fano, Effects of configuration interaction on intensities and phase shifts, *Physical Review* **124**, 1866 (1961).
- [20] R. Adler, A study of locking phenomena in oscillators, *Proceedings of the IRE* **34**, 351 (1946).
- [21] B. van der Pol and J. van der Mark, Frequency demultiplication, *Nature* **120**, 363 (1927).
- [22] S. Shapiro, Josephson currents in superconducting tunneling: The effect of microwaves and other observations, *Physical Review Letters* **11**, 80 (1963).
- [23] K. L. Turner, S. A. Miller, P. G. Hartwell, N. C. MacDonald, S. H. Strogatz, and S. G. Adams, Five parametric resonances in a microelectromechanical system, *Nature* **396**, 149 (1998).
- [24] J. Rajchenbach, A. Leroux, and D. Clamond, New standing solitary waves in water, *Nature Communications* **12**, 1 (2021).
- [25] J. Zhang, P. W. Hess, A. Kyprianidis, P. Becker, A. Lee, J. Smith, G. Pagano, I.-D. Potirniche, A. C. Potter, A. Vishwanath, N. Y. Yao, and C. Monroe, Observation of a discrete time crystal, *Nature* **543**, 217 (2017).
- [26] S. Choi, J. Choi, R. Landig, G. Kucsko, H. Zhou, J. Isoya, F. Jelezko, S. Onoda, H. Sumiya, V. Khemani, C. von Keyserlingk, N. Y. Yao, E. Demler, and M. D. Lukin, Observation of discrete time-crystalline order in a disordered dipolar many-body system, *Nature* **543**, 221 (2017).
- [27] R. Benzi, G. Parisi, A. Sutera, and A. Vulpiani, A theory of stochastic resonance in climatic change, *SIAM Journal on Applied Mathematics* **43**, 565 (1983).